oxidase and are transferred to other members of chain as they transfer the electron to the next component (Figure 11.26).

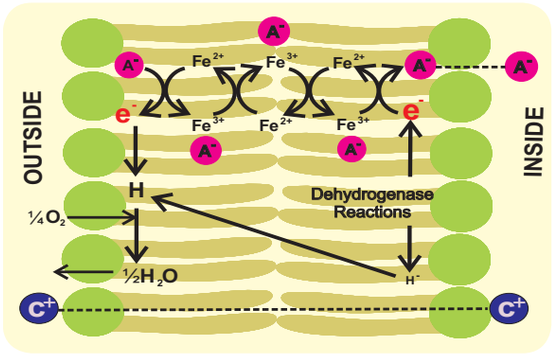


Figure 11.26: Cytochrome Pump theory

The theory assumes that cations (C^+) move passively along the electrical gradient created by the accumulation of anions (A^-) at the inner surface of the membrane.

Main defects of the above theory are:

(i) Cations also induce respiration.

(ii) Fails to explain the selective uptake of ions.

(iii) It explains absorption of anions only.

2. Bennet-Clark's Protein-Lecithin Theory: In 1956, **Bennet-Clark** proposed that the carrier could be a protein associated with **phosphatide** called as **lecithin**. The carrier is **amphoteric** (the ability to act either as an acid or a base) and hence both cations and anions combine with it to form **Lecithin-ion complex** in the membrane. Inside the membrane, Lecithin-ion complex is broken down into **phosphatidic acid** and **choline** along with the liberation of ions. Lecithin again gets regenerated from *phosphatidic acid* and *choline* in the presence of the enzyme *choline acetylase* and *choline esterase* (Figure 11.27). ATP is required for regeneration of lecithin.

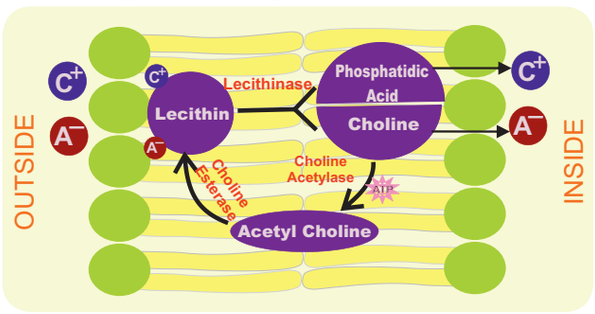


Figure 11.27: Protein-Lecithin theory

11.8.3 Donnan equilibrium

Within the cell, some of the ions never diffuse out through the membrane. They are trapped within the cell and are called fixed ions. But they must be balanced by the ions of opposite charge. Assuming that a concentration of fixed anions is present inside the membrane, more cations would be absorbed in addition to the normal exchange to maintain the equilibrium. Therefore, the cation concentration would be greater in the internal than in the external solution. This electrical balance or equilibrium controlled by electrical as well as diffusion phenomenon is known as the **Donnan equilibrium**.

Summary

There are two types of transports namely short and long distance in plants to translocate sap and solutes. Based on energy requirement, the transport may either be passive or active. The process of diffusion, facilitated diffusion, imbibition and osmosis are driven by concentration gradient like a ball rolling down to a slope and hence, no energy is needed. The water absorbed (either active or passive) from the soil by root hairs must reach the xylem for further transportation. There are three possible routes to reach the xylem from root hairs. They are i) apoplast ii) symplast and/or iii) transmembrane. Various theories explain the path of sap in the xylem and Dixon's Cohesion-tension theory is the most accepted one. Transpiration is mostly carried out by stomata, which has guard cells. The general mechanism of stomatal movement is based on entry and exit of water molecules in guard cells. Many theories are there to explain how water enters and exits from guard cells. The theory of potassium transport enumerates two different reactions separately run for opening and closing of stomata. Contrary to ascent of sap by xylem in an upward direction, the path of solute which consists of the photosynthetic products is always in phloem and translocate multidirectional. The point of origin of translocation is photosynthetic leaves which

XIth_Botany_Chapter_11_EM.indd 256 05-01-2023 18:29:12